HPP-IR: Hierarchical Progressive Prompting for All-in-One Weather-Degraded Image Restoration

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***Abstract*—All-in-one image restoration aims to recover high-quality images from diverse weather degradations using a single model. However, existing prompt-based methods often rely on single-level prompting and overlook the severe dilution of degradation cues during deep encoding, leading to feature confusion. Moreover, their training paradigms lack targeted designs for all-in-one restoration, resulting in suboptimal performance. To address this, we propose HPP-IR, a hierarchical progressive two-level prompting architecture. First, a Degradation-Feature Prompt Module (DFPM) operates at the encoder stage to extract and preserve original degradation priors, effectively preventing feature dilution and accurately disentangling weather effects from intrinsic content. Second, a Cross-Layer Prompt Component (CLPC) converts these priors into context-aware prompts within encoder-decoder skip connections, enhancing the specificity of feature transfer. Furthermore, we integrate the Pearson Correlation Coefficient into the loss function to enforce intrinsic pixel-intensity correlations, strictly preserving natural structures and visual consistency. This work represents an early systematic exploration into a two-level collaborative prompting paradigm that synergistically integrates prompt mechanisms at both encoder stages and cross-layer skip connections for all-in-one image restoration. Extensive experiments demonstrate that our method achieves superior performance across diverse weather degradation scenarios with a lightweight design, enabling efficient deployment in real-time visual systems.**

***Index Terms*—All-in-one Image Restoration; Prompt Learning; Progressive Prompting.**

I. Introduction

HIGH-QUALITY and clear visual information constitutes the fundamental prerequisite and essential guarantee for computer vision tasks and video systems. However, in outdoor vision scenarios, image quality is frequently compromised by adverse weather conditions such as haze, rain, and snow, which severely degrade image usability. In recent years, single-degradation restoration methods have achieved remarkable success in dedicated tasks, including dehazing [1], [2], deraining [3], raindrop removal [4], desnowing [5] and other specific restoration tasks [6]–[9]. For instance, recent advancements have explored cross-level feature interactions for remote sensing dehazing [10], and introduced optimal transport theory to facilitate unpaired photo-realistic dehazing [11] as well as text-guided rain removal [12]. Nevertheless, these highly specialized methods exhibit poor generalization and cannot handle multiple degradation types simultaneously. Consequently, developing a high-performance all-in-one restoration model that can effectively counteract complex atmospheric weather degradations while meeting real-world deployment requirements is of paramount importance to the reliability of vision-based applications.

Early all-in-one image restoration methods primarily relied on sophisticated network architectures. Task-specific encoder–decoder designs [13]–[15] trained separate networks for each degradation type and achieved satisfactory performance. However, these schemes required prior knowledge of the degradation category to select the appropriate component, thereby lacking flexibility and practicality. In addition, the requirement for multiple networks imposed high computational and resource demands. Subsequent approaches—such as shared-decoder architectures with multiple heads [16], shared-backbone architectures with multiple decoders [17], and unified encoder–decoder frameworks [18]–[23], addressed diverse degradations within a single framework, eliminating the need for degradation-type priors and offering improved flexibility and efficiency. However, these prompt-free methods still struggle to adapt dynamically to the specific degradation characteristics of input images and exhibit limited generalization when confronted with unseen degradation patterns, thereby failing to fully meet the practical deployment requirements.

Meanwhile, the rise of large language models (LLMs) has led to a revolutionary breakthrough in the prompt-learning paradigm, which seeks to exploit the intrinsic knowledge of pretrained language models by providing instructions or relevant information, thereby significantly reducing downstream task adaptation costs and enhancing generalization capabilities. This paradigm has rapidly migrated to the vision domain: CLIP [24] pioneered vision–language alignment, laying the foundation for cross-modal prompting, while subsequent text-to-image models [25]–[27] have enabled high-quality image generation driven solely by textual prompts. Recently, a number of researchers have applied large model-based prompting to image restoration [28]–[32], achieving controllable and dynamic guidance of the restoration process to effectively enhance adaptability and controllability under complex or unknown degradations. However, these methods demand substantial hardware resources and exhibit slow inference speeds, which constrain their deployment.

In recent years, numerous studies have investigated lightweight, prompt-based, all-in-one image restoration models. Some methods directly concatenate visual prompts with degraded images as the input [33], [34], providing the model with highly intuitive spatial guidance. A few studies [35], [36] integrate prompts—generated by lightweight plug-and-play

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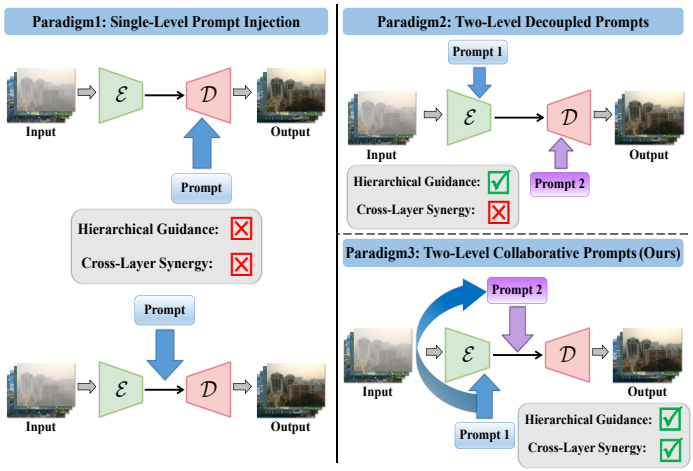


Fig. 1. Comparison of visual prompt injection paradigms in image restoration.

prompt modules—into encoder stages, delivering prior knowledge to guide feature extraction. Other studies [37]–[39] inject encoded prompt vectors into intermediate feature layers between the encoder and decoder, enabling fine-grained dynamic modulation of feature responses. Numerous methods [40]–[43] deploy prompts along the hierarchical evolution path within the decoding stage, performing task representation or variable updates in residual form to guide restoration. Researchers have also explored embedding prompts into skip connections of encoder–decoder architectures [44], [45], allowing for the dynamic recalibration of cross-scale features and enabling a single network to adapt to multiple restoration tasks. Beyond these single-stage deployments, to provide richer contextual guidance, recent works have also explored two-level decoupled prompt designs [29], [30], attempting to simultaneously inject prompts at multiple disconnected stages within the network hierarchy.

The aforementioned approaches can be structurally abstracted into three prompt injection paradigms (Fig. 1). Most models adopt Paradigm 1 (Single-Level Prompt Injection) at an isolated stage. However, this strictly localized modulation restricts comprehensive guidance, limiting its capacity to dynamically modulate hierarchical features. To provide richer context, recent works explore Paradigm 2 (Two-Level Decoupled Prompts). Yet, its physically decoupled design precludes cross-level interaction, which easily induces conflicting optimization directions and compromises feature synergy. However, these existing paradigms overlook a bottleneck in deep networks: vulnerability to feature dilution. Feature dilution refers to the phenomenon where high-frequency weather degradation cues are progressively compressed and overwhelmed by dominant background scene semantics as the encoder deepens, leading to severe feature confusion during decoding. To overcome these limitations, we propose Paradigm 3 (Two-Level Collaborative Prompts).

By deploying interactive prompts at both the encoder and skip connections—where the encoder-derived prompt explicitly conditions the cross-layer prompt—our synergistic architecture precisely aligns features, prevents optimization conflicts, and fully exploits the potential of multi-level prompting for superior all-in-one restoration. While our collaborative prompting architecture effectively optimizes feature-level representations, existing loss-function designs often focus solely on pixel-level intensity similarity, neglecting the disruption of intrinsic pixel-intensity correlations caused by weather degradations, which leads to restoration results with suboptimal natural structure and detail fidelity.

To address these issues, this study proposes a progressive two-level prompting architecture—HPP-IR, which employs comprehensive prompts to enhance image restoration, adaptively guides critical feature transfer, and reinforces the recovery of intrinsic pixel correlations, thereby significantly improving model efficiency and restoration quality without sacrificing performance. The main contributions of this study are:

1) Degradation-Feature Prompt Module (DFPM): To prevent the severe dilution of degradation information and facilitate the accurate disentanglement of weather effects from intrinsic image content as the encoder deepens, we design the DFPM. It explicitly extracts degradation-specific priors from the degraded image and supplies multidimensional prompts to different encoder layers, thereby enhancing the model's understanding of the image structure and significantly improving the accuracy of feature decoupling at every encoder layer.

2) Cross-Layer Prompt Component (CLPC): To enhance the specificity of encoder–decoder feature transfer, we propose CLPC, in which the Prompt Composition Unit (PCU) produces degradation-related feature guidance conditioned on each input image, and the Prompt Fusion Unit (PFU) fuses these prompts with encoder output features. This selectively amplifies restoration-relevant features while suppressing irrelevant noise, enhancing the model's robustness under complex weather conditions.

3) Pearson Correlation Coefficient (PCC): To address the degradation-induced disruption of pixel-intensity correlations overlooked in the existing loss functions, We systematically introduce and validate the applicability of the PCC loss within the context of all-in-one image restoration. This preserves natural structure and visual consistency, effectively mitigating structural distortion and over-smoothing.

II. Related Work

A. Early All-in-One Image Restoration

Early all-in-one image restoration approaches aimed to simultaneously handle multiple degradation types through sophisticated network architectures, establishing the foundation for subsequent advancements. Li et al. [16] leveraged neural architecture search and physics-driven tensor operations to design a multi-head shared-decoder architecture that unifies adverse-weather restoration without requiring prior knowledge of degradation types; BIDeN [17] adopts a single-encoder–multi-decoder topology with mixed inputs and a shared backbone for feature extraction, yet it reintroduces the complexity of multiple decoders and demands degradation-type supervision.

Unified encoder–decoder frameworks dispense with separate heads or tails: AirNet [18] captures multi-degradation

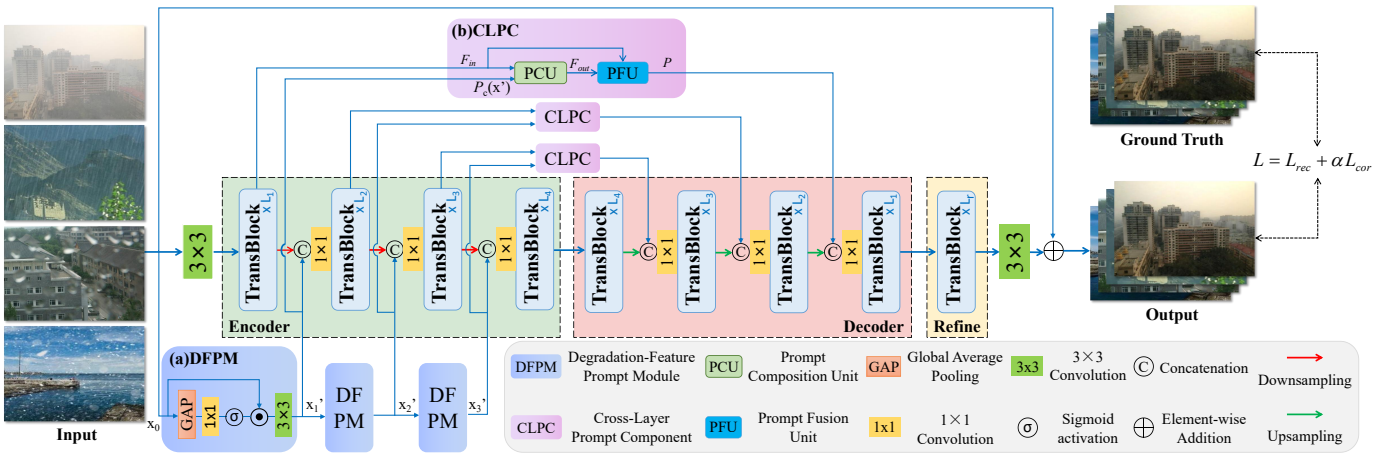


Fig. 2. Overall architecture of the proposed HPP-IR network.

representations via contrastive learning and performs adaptive restoration with deformable convolutions and spatial feature transforms; TANet [21] introduces a triplet-attention mechanism that combines local pixel attention, global stripe attention, and global distribution attention to effectively restore images under adverse weather conditions; AdaIR [44] employs frequency-mining and frequency-modulation modules to extract and interactively optimize diverse frequency components from input images, thereby enhancing the performance of image restoration.

While these methods have significantly advanced all-in-one image restoration, they have not incorporated prompt learning, lacking a degradation-aware dynamic guidance mechanism and failing to adaptively disentangle mixed degradation features from background content, which limits the overall robustness of vision systems in complex environments.

B. Large-Model-Prompt All-in-One Image Restoration

Methods in this category leverage the powerful feature-extraction capabilities of large-scale pre-trained models and have achieved significant progress.As a representative work leveraging large diffusion models, PromptGIP [28] randomly masks portions of an image and instructs the model to reconstruct the masked patches by leveraging unmasked regions; during inference, input–output pairs are formatted as task prompts to guide the restoration process.

Natural-language instructions or multi-modal prompts that combine visual and textual cues improve adaptability to unknown degradations: Conde et al. proposed InstructIR [29], the first human-instruction-driven restoration model guided by free-form language commands, which injects instruction embeddings into both encoder and decoder blocks; TextPromptIR [30] employs a fine-tuned BERT to generate semantic prompts and independently injects them into both the encoder and decoder stages, bridging the gap between textual prompts and visual features via deep multi-head transpose attention and gated convolutions; Instruct-IPT [31] adopted weight modulation and synchronous training to efficiently learn task-specific knowledge while maintaining universality.

Pre-trained model priors harness frozen vision-language models such as CLIP, DINO [46], and DINO-v2 [47] to predict high-quality feature embeddings by leveraging semantic alignment between vision and language. Perceive-IR [48] employs CLIP for prompt learning to distinguish image quality levels and integrates DINO-v2 features to guide restoration; DA-CLIP [49] introduces a degradation-aware vision–language model that adaptively adjusts the frozen CLIP image encoder via an auxiliary controller to output content embeddings aligned with clean-image features.

These approaches benefit from rich prior knowledge but face substantial computational overhead, posing challenges to the deployment.

C. Small-Model-Prompt All-in-One Image Restoration

Small-model prompting methods remain lightweight while enhancing model flexibility and generalization. ProRes [33] introduces targeted visual prompts to the input image to generate a prompt image, only these randomly initialized prompt parameters require fine-tuning for new restoration tasks or datasets; ConStyle v2 [36] proposes a lightweight plug-and-play prompter that, via two-stage training, injects clean visual prompts generated by the lightweight ConStyle network into any U-Net encoder to provide degradation-agnostic priors; PIP [37] embeds degradation-aware and base-restoration prompts into encoder-decoder skip connections, progressively regulating features through ”prompt-to-prompt” and ”prompt-to-feature” interactions; DFPIR [39] introduces a Degradation-Guided Perturbation Block in the same location, perturbing encoder features with textual prompts and concatenating it with decoder features to achieve explicit degradation-aware guidance; DPPD [40] first anchors degradation features with orthogonal prototypes, then formulates prompts as a sampleable Gaussian mixture distribution, and injects dynamic prompts via cross-attention into the decoder; PromptIR [41] adopts a single encoder-decoder architecture and progressively injects learnable visual prompts at multiple decoding stages to implicitly infer degradation conditions, modulating feature responses via channel-wise multiplication and adaptive gating mechanisms; Up-Restorer [42] inserts Prompt Blocks along the iterative denoising steps of diffusion model, dynamically generating and updating physical variables (transmission map, atmospheric light, noise map, etc.) at each iterative step to steer the diffusion process. MEASNet [50] introduces a multi-

expert adaptive selection mechanism, leveraging task-specific prompts at the MoE routing gating to balance diverse tasks within a single framework. additionally, adaptive degradation-aware self-prompting models [51] dynamically perceive and restore weather-degraded images without relying on external task supervision.

However, existing methods exhibit limitations in prompt generation, propagation, and loss supervision. Furthermore, current architectures largely fall into either single-level prompt injection or two-level decoupled prompting paradigms, both of which struggle with suboptimal feature synergy and vulnerability to deep-layer feature dilution.

III. Proposed Method

This paper presents HPP-IR, a progressive two-level prompting architecture for all-in-one image restoration, which enhances the model's capability to restore images degraded by complex weather conditions through layer-wise progressive prompting. The overall framework is illustrated in Fig. 2. Built upon the classical U-Net backbone, our method employs a dual-level prompting mechanism: **(1)** The first-level Degradation-Feature Prompt Module (DFPM) operates in the encoder stage with a dual-branch design, explicitly extracting original degradation priors from degraded images to provide foundational prompt information for subsequent encoder layers. This prevents severe dilution of degradation information as the encoder deepens and facilitates accurate disentanglement of weather effects from intrinsic image content. **(2)** With original degradation priors adaptively preserved and features accurately disentangled in the encoder, the challenge thus shifts to effectively guiding the transfer of these refined features to the decoder for task-beneficial reconstruction. To address this, the second-level Cross-Layer Prompt Component (CLPC) is embedded within encoder-decoder skip connections, dynamically generating and fusing feature prompts to enhance feature transfer specificity and model robustness under adverse weather conditions. **(3)** Despite these feature-level advances, dedicated supervision at the pixel-intensity level is still required to enforce structural consistency. Therefore, we develop a loss function that incorporates the Pearson Correlation Coefficient (PCC). By directly constraining intrinsic pixel intensity correlations alongside pixel-level similarity, the PCC loss complements the preceding feature-level modules, collectively ensuring a holistic recovery from local details to global structures.

To construct a lightweight architecture, we adopt the classical lightweight self-attention mechanism and feed-forward network (FFN) from the baseline Restormer [52] in all encoder, decoder, and refinement stages. The following sections sequentially detail the structure of the first-level degradation-feature prompt module, the second-level prompt component along with its internal sub-modules, and the design of the loss function.

A. Degradation-Feature Prompt Module

In image restoration, complex weather conditions such as rain, snow, and haze introduce diverse degradation patterns that severely obscure the underlying image content. As the network depth increases in traditional encoder architectures, the repeated application of downsampling and feature transformation operations inevitably leads to the dilution of high-frequency and fine-grained degradation cues. Consequently, the deep layers of the encoder tend to prioritize high-level semantic representations while losing track of the specific weather degradation characteristics. Furthermore, because certain weather degradations share visual similarities with intrinsic image textures (e.g., snow particles resembling white textured walls), the loss of these explicit degradation cues often causes deep network layers to entangle weather effects with genuine semantic content, resulting in feature confusion and suboptimal restoration.

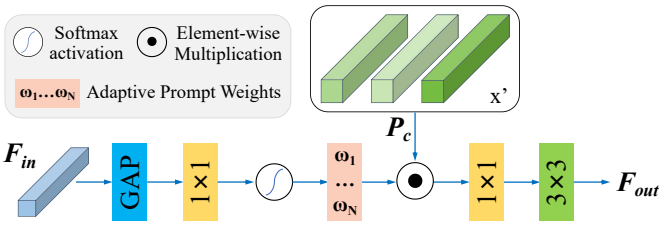


Fig. 3. Prompt Composition Unit within the Cross-Layer Prompt Component.

To tackle these issues and prevent the dilution of degradation information, we propose a simple yet effective encoder-stage structure termed the Degradation-Feature Prompt Module (DFPM). Since the first TransBlock directly acquires intact degradation cues from the input image, the DFPM is strategically deployed from the second stage onward to avoid computational redundancy and shallow-level interference. It progressively embeds lightweight-modulated degradation prompts into every deeper encoder layer via coarse skip connections in a residual fashion, avoiding scale or semantic mismatches that may result from direct concatenation. The core idea is that one path preserves the conventional hierarchical feature flow, while the other establishes a direct connection to the raw input. This dual-path mechanism enables each deep layer to simultaneously access both abstract semantic features from above and concrete degradation priors from the original image, preventing information dilution and enabling the accurate decoupling of weather-related degradations from background content. As shown in Fig. 2 (a), DFPM consists of a sequence of operations expressed by:

$$x' = \mathrm{Conv}_{3\times3}\Big(x_0 \odot \mathrm{Sigmoid}\big(\mathrm{Conv}(\mathrm{GAP}(x_0))\big)\Big), \quad (1)$$

where GAP is Global Average Pooling, Conv indicates a 1×1 convolution, and $\mathrm{Conv}_{3\times3}$ denotes a 3×3 conventional convolution. Weather degradations exhibit two dominant patterns: local particles (e.g., rain streaks or snowflakes) manifest as spatial texture variations, whereas global atmospheric effects (e.g., haze or illumination shift) appear as channel-wise style statistics. Convolution's inherently local receptive field, however, tends to lose global context at deeper layers. We therefore start with GAP to compress each channel into a single scalar, preserving overall degradation strength while drastically reducing spatial dimension, thus supplying

lightweight yet critical background information for subsequent modulation. To align the global-pooled scalar with the original channel dimension and refine its representation, a 1×1 convolution expands it to match the input channel space, followed by Sigmoid to produce per-channel weights in [0, 1]. These weights adaptively amplify weather-relevant channels and suppress irrelevant noise, realizing efficient channel-wise dynamic modulation. However, relying solely on this SE-like channel recalibration would discard the local spatial priors that are crucial for modeling fine-grained particle degradations (e.g., rain and snow). Therefore, unlike standard channel attention mechanisms, we intentionally introduce a 3×3 standard convolution to adaptively refine the channel-wise modulation weights and fuse them with the spatial context. This crucial step unifies the global degradation strength with fine-grained local textures. Chained together, these operations extract and fuse multi-scale raw and encoder features with minimal parameters.

Moreover, unlike standard intra-layer attention modules that only operate within a single depth level, DFPM establishes a cross-layer parallel pathway. From the second stage onward, the modulated raw features are progressively injected into each layer, providing degradation-agnostic references and focusing the network on learning the weather degradation residual. Experiments demonstrate that this lightweight design can effectively enhance encoder flexibility, precisely disentangle diverse weather degradations, prevent feature dilution, and maintain highly accurate content representations, thereby delivering superior restoration under complex atmospheric weather conditions.

B. Cross-Layer Prompt Component

Although the DFPM, serving as the first-level prompt, has already captured and preserved the original degradation priors in parallel during the encoding stage, providing a fundamental basis for feature disentanglement at each layer. However, direct skip-connection methods traditionally transfer features without explicitly distinguishing between spatially-variant local degradations and globally-distributed atmospheric effects, making them prone to information confusion during the decoding phase. Therefore, to provide the required fine-grained modulation of encoder features before their integration into the decoder, we embed a Cross-Layer Prompt Component (CLPC) within the skip connections as a second-level prompt. As shown in Fig. 2(b), this ensures the refined priors accurately guide the final restoration.

This component interacts with the encoder's output features, dynamically converting the preserved degradation priors into precise prompts that are deeply coupled with the decoder features, thereby adaptively guiding the decoder's generation process. The CLPC comprises two main sub-modules: the Prompt Composition Unit (PCU) and the Prompt Fusion Unit (PFU). The PCU is responsible for dynamically generating degradation-type-related feature guidance based on the encoder's output features, providing adaptive guidance for the subsequent interaction. The PFU then fuses these dynamic prompts with the encoder features, selectively enhancing

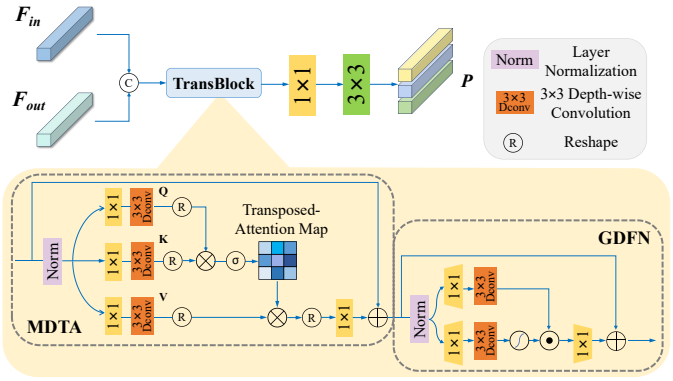


Fig. 4. Prompt Fusion Unit within the Cross-Layer Prompt Component.

restoration-task-related features while suppressing irrelevant noise information. The corresponding process is given in Eq.2:

$$P = \mathrm{PFU}\Big(\mathrm{PCU}\big(P_c,\ F_{in}\big),\ F_{in}\Big). \tag{2}$$

The specific operation procedures of the PFU and PCU will be introduced in the following.

1) Prompt Composition Unit: The original degradation priors preserved by the DFPM are encapsulated as learnable prompt components P_c and fed into the PCU. Static feature interaction methods cannot generate input-specific prompts and thus lack adaptability, usually resulting in inferior performance under diverse input conditions. Therefore, this paper employs the PCU to dynamically predict attention-based weights conditioned on the input features and applies these weights to the prompt components to generate input-conditioned prompts as shown in Fig. 3.

To generate prompt weights from the input features, the PCU first applies GAP along the spatial dimension to obtain a feature vector $v \in \mathbb{R}^C$, compresses it via a channel-downscaling convolution, and then uses a softmax operation to produce the normalized prompt weights $\omega \in \mathbb{R}^N$. These weights are subsequently used to modulate the prompt content P_c, followed by a 1×1 convolution and a 3×3 convolution to refine the resulting features. Overall, the PCU pipeline is summarized as follows:

$$\begin{aligned} F_{\text{out}} &= \mathrm{Conv}_{3\times3}\Big(\mathrm{Conv}\big(\sum_{i=1}^{N}\omega_i P_c\big)\Big), \\ \omega_i &= \mathrm{Softmax}\big(\mathrm{Conv}(\mathrm{GAP}(F_{in}))\big). \end{aligned} \tag{3}$$

2) Prompt Fusion Unit: As shown in Fig. 4, within the PFU, we first concatenate the encoder features F_{in} and the dynamic prompts F_{out} along the channel dimension. The concatenated representation is then fed into a Transformer block, which leverages the degradation information encoded in the prompts to adaptively transform the input features. The Multi-Dconv Head Transposed Attention (MDTA) in the TransBlock employs cross-channel self-attention instead of spatial-dimension operations, achieving linear complexity. This is followed by a Gated-Dconv Feed-Forward Network (GDFN), which transforms features through a controlled gating mechanism: suppressing residual degradation information

TABLE I
QUANTITATIVE COMPARISON OF RESTORATION RESULTS OF IMAGE RESTORATION METHODS ON THE PROPOSED DATASETS

Methods	SOTS		Rain1400		Raindrop		Snow100K		Average		Coupled (Zero-shot)	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
MPRNet [53]	26.64	0.9428	32.44	0.9366	29.39	0.9000	28.53	0.9006	29.25	0.9200	16.35	0.5725
AirNet [18]	27.83	0.9503	24.73	0.7890	26.17	0.8674	26.04	0.8789	26.19	0.8714	16.07	0.5832
TransWeather [20]	25.40	0.9332	30.14	0.8968	28.84	0.9420	28.84	0.9260	28.31	0.9245	15.10	0.5666
WGWS-Net [54]	27.97	0.9213	32.49	0.9324	31.35	0.9228	27.49	0.8870	29.83	0.9159	25.15	0.8933
PromptIR [41]	24.43	0.8870	30.82	0.9131	30.06	0.9076	25.88	0.8658	27.80	0.8934	13.67	0.5720
GridFormer [55]	26.86	0.9070	31.90	0.9033	32.23	0.9334	29.36	0.9083	30.09	0.9130	16.63	0.5851
Histoformer [56]	28.69	0.9017	32.21	0.9194	30.38	0.8908	29.70	0.9129	30.25	0.9062	26.53	0.8830
MEASNet [50]	28.39	0.9687	31.29	0.9167	29.15	0.8818	30.37	0.9139	29.80	0.9203	15.15	0.5920
AdaIR [44]	29.86	0.9747	31.38	0.9182	30.74	0.9155	29.34	0.8869	30.33	0.9238	14.42	0.5793
AWRaCLe [57]	27.22	0.9280	31.48	0.9172	29.45	0.9075	29.16	0.8990	29.33	0.9129	13.59	0.5814
MPMF-Net [58]	27.24	0.9298	29.17	0.8852	26.04	0.8766	26.26	0.8652	27.18	0.8892	16.90	0.5854
BaryIR [59]	27.80	0.9339	29.95	0.8972	26.76	0.8685	26.51	0.8684	27.76	0.8920	16.56	0.5926
HPP-IR(Ours)	29.32	0.9321	33.19	0.9510	31.01	0.9465	30.15	0.9355	30.92	0.9413	28.95	0.8988

while allowing only restoration-relevant features to propagate. The operations of the PFU are mathematically formulated as follows:

$$P = \mathrm{Conv}_{3\times3}\Big(\mathrm{Conv}\big(\mathrm{GDFN}\big(\mathrm{MDTA}\big[F_{\mathrm{in}} \copyright F_{\mathrm{out}}\big]\big)\big)\Big), \quad (4)$$

where © denotes channel-wise concatenation. The core objective of the CLPC is to generate input-conditioned prompts that supply the decoder with useful contextual information, guiding the restoration network to remove degradation patterns from the input image more effectively.

In summary, embedding the prompt component into the encoder–decoder skip connections not only explicitly distinguishes spatially-variant local degradation particles from globally-distributed atmospheric effects but also enhances the specificity and adaptability of feature transfer. This avoids potential feature confusion inherent in traditional skip connections and significantly boosts image restoration performance under complex weather conditions. At each decoder level, the prompt component implicitly enriches the decoder features with context-aware restorative guidance, further improving restoration performance in adverse weather. Experimental results demonstrate that this design enhances both the targeting and efficacy of encoder–decoder feature transfer, offering a novel solution for image-restoration tasks.

C. Loss Function

While establishing a hierarchical progressive prompting architecture via DFPM and CLPC is crucial for maximizing the model's discriminative and restorative capabilities at the feature level, we observe that ensuring intrinsic structural consistency at the pixel-intensity level requires dedicated supervision, which is lacking in existing loss functions.

Motivated by this observation, we introduce and analyze a comprehensive loss function consisting of two parts: the reconstruction loss $\mathcal{L}_{rec}$ and correlation loss $\mathcal{L}_{cor}$, tailored to the characteristics of weather-degraded image restoration. For the reconstruction loss, we adopt the classic $\mathcal{L}_1$ loss to impose pixel-level constraints between the processed image and the ground-truth image, as shown in Eq. 5, where I_{gt} and $I_{restored}$ denote the ground-truth image and the processed image, respectively:

$$\mathcal{L}_{rec} = \left\| I_{restored} - I_{gt} \right\|_1. \quad (5)$$

However, weather degradations not only degrade visual quality but also drastically alter pixel-intensity distributions—global brightness shifts caused by haze [60], local blurring introduced by raindrops or snowflakes—thereby destroying the original intensity relationships and local structures. While commonly used structural constraints like SSIM evaluate local windows and struggle with severe, globally distributed intensity shifts, and perceptual losses rely on cumbersome pre-trained networks that substantially increase memory footprint without guaranteeing structural fidelity, they both fundamentally ignore the severe disruption of natural pixel correlations. This oversight often leads to structural distortion and detail loss in the restored results.

To address this limitation, we adapt and introduce a correlation-aware constraint alongside the classic $\mathcal{L}_1$ loss. The Pearson Correlation Coefficient (PCC), a statistical measure of linear correlation between two variables ranging from -1 to +1, with values closer to 1 indicating stronger consistency in intensity trends, effectively characterizes the intrinsic ordering and distribution of pixel intensities across local or global regions . By supervising the network with the PCC, we enforce consistent intensity trends between the restored and ground-truth images in multi-degradation tasks. This approach preserves natural structure and fine details without increasing network complexity. The correlation loss is formulated as:

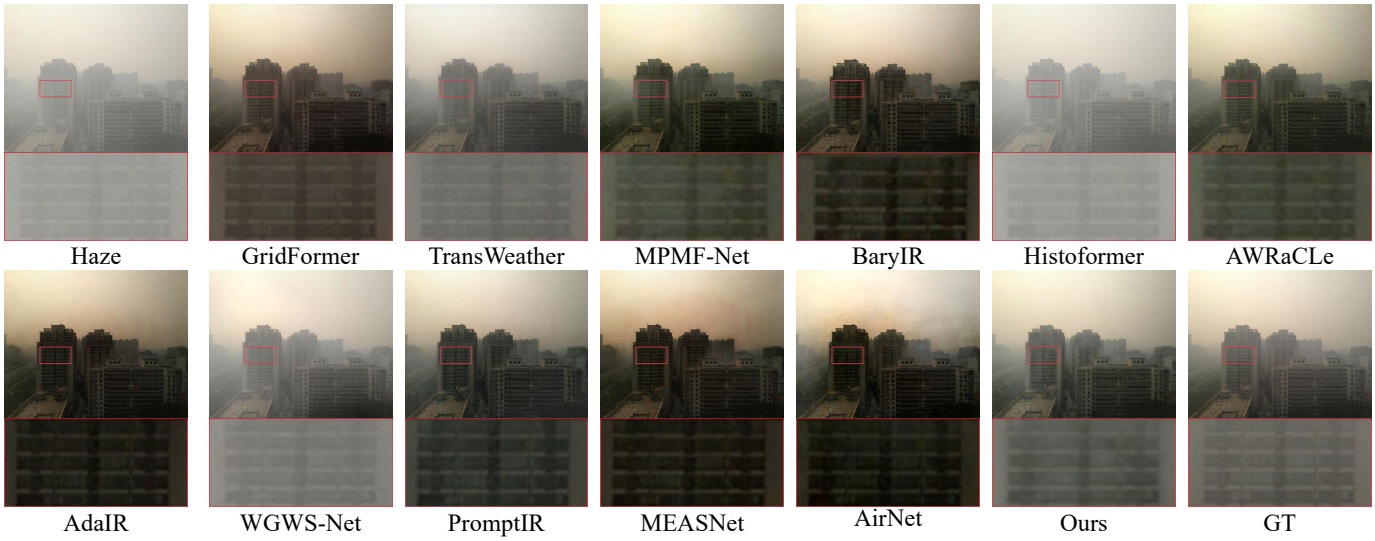


Fig. 5. Visual comparisons for dehazing on SOTS dataset.

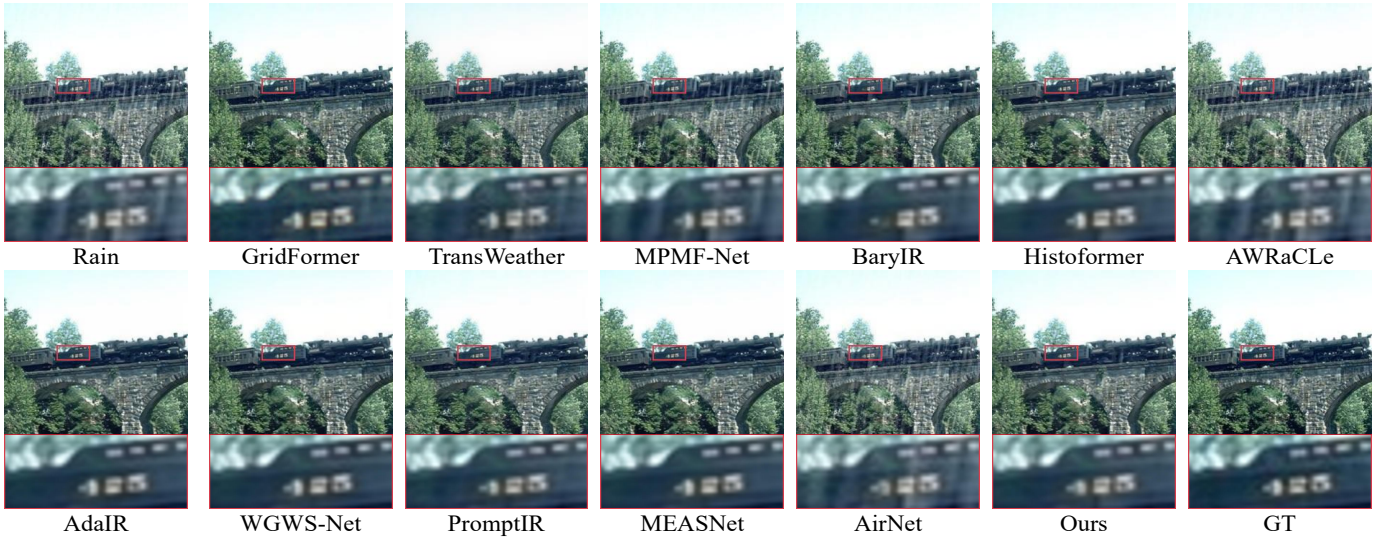


Fig. 6. Visual comparisons for deraining on Rain1400 dataset.

$$\rho\left(I^{restored},\, I^{gt}\right) = \frac{\sum_{i=1}^{3HW}\left(I_i^{restored} - \bar{I}^{restored}\right)\left(I_i^{gt} - \bar{I}^{gt}\right)}{3HW\,\sigma(I^{restored})\,\sigma(I^{gt})}, \tag{6}$$

where $I_i^{\{\cdot\}}$ denotes the i-th pixel of the image, $\bar{I}^{\{\cdot\}}$ and $\sigma\left(I^{\{\cdot\}}\right)$ are the mean and standard deviation of the image sequence, respectively. Since ρ intrinsically falls within $[-1, 1]$, we formulate the correlation loss to penalize structural deviations as:

$$\mathcal{L}_{cor} = \frac{1 - \rho\left(I^{restored},\, I^{gt}\right)}{2}. \tag{7}$$

The closer $\mathcal{L}_{cor}$ is to 0, the more accurately the restored image follows the intensity ordering and distribution of the ground-truth image, and the degraded pixel relationships caused by weather are restored to their intrinsic patterns.

The overall loss function is defined as follows, where α is the weight of the correlation loss:

$$L = L_{rec} + \alpha L_{cor}. \tag{8}$$

In this way, the PCC-based loss complements the feature-level optimizations of the preceding modules, working in concert to ensure comprehensive recovery quality from local features to global structural fidelity.

IV. Experiments

A. Experimental settings

This study is implemented using Python and the PyTorch framework on an Ubuntu operating system, accelerated by an NVIDIA GeForce RTX 3090 GPU with CUDA for all experiments. The network architecture employs four encoder-decoder layers, with [4, 4, 6, 8] basic modules per layer and attention heads increasing as [1, 2, 4, 8] with network depth. We implement a progressive training strategy with batch sizes [4, 2, 2, 1, 1] and patch sizes [128, 160, 160, 256, 256] over five stages, enabling coarse-to-fine learning. The total 300K iterations follow a cosine-annealing-restart schedule with initial learning rate fixed at 3×10^{-4} and gradually decaying to 10^{-6} over the subsequent 208K iterations, avoiding loss oscillations and improving convergence stability. The AdamW optimizer employs β_1 = 0.9, β_2 = 0.999, and a weight decay

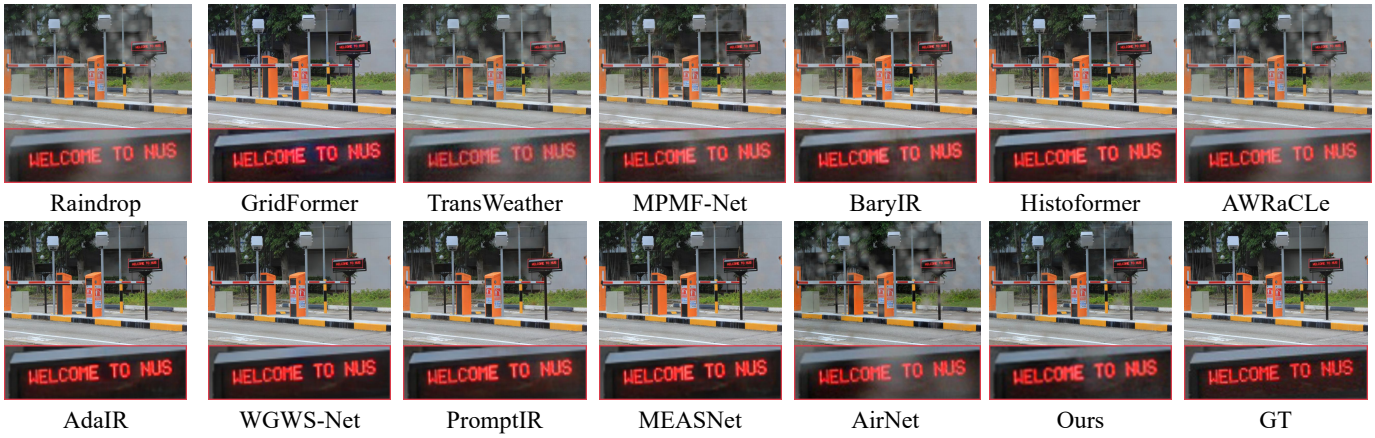


Fig. 7. Visual comparisons for raindrop removal on Raindrop dataset.

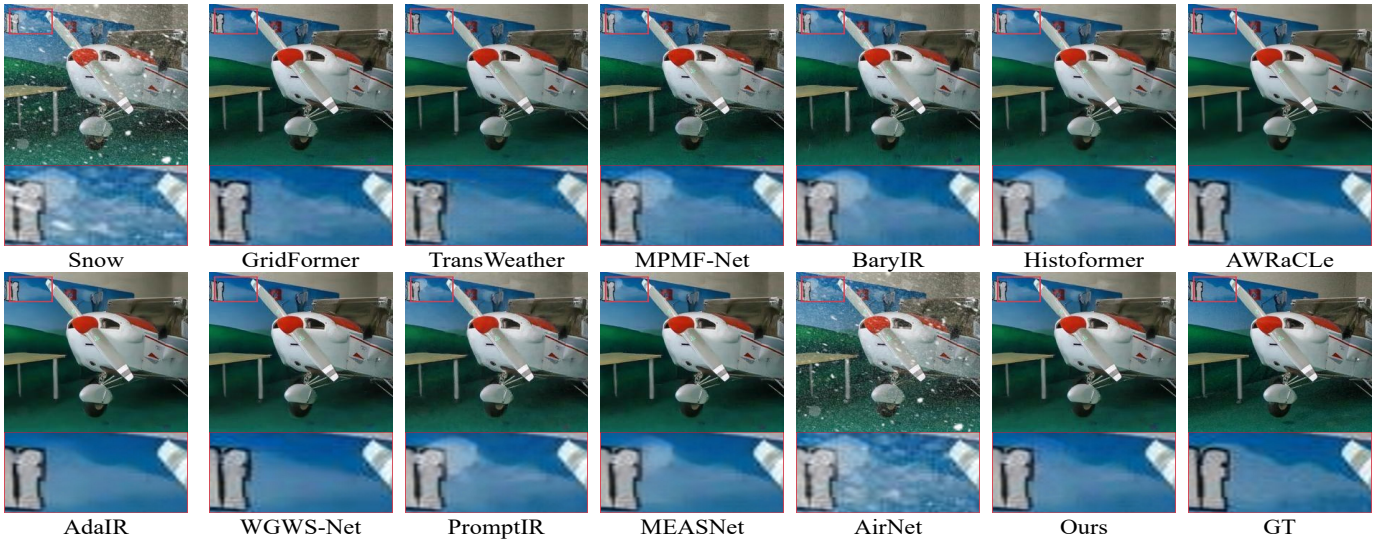


Fig. 8. Visual comparisons for desnowing on Snow100K dataset.

coefficient of 10^{-4} to prevent model overfitting. Furthermore, the balancing hyperparameter α for the correlation loss $\mathcal{L}_{cor}$ is set to 1, based on the optimal configuration identified in our ablation study.

For image datasets, RESIDE (OTS for training, 72,135 images; SOTS for testing, 500 images) is used for dehazing. Rain-streak removal employs Rain1400 (12,600 training; 1,400 testing). Raindrop removal uses Raindrop (861 training; 58 test_a images). Desnowing adopts Snow100K (50,000 training: 16,643 light, 16,622 medium, 16,735 heavy snow; 16,801 SNOW100K-L test images). Generalization is validated on a rain-haze coupled dataset, which is strictly excluded from the training phase of all models to serve solely for zero-shot assessment.

To ensure absolute fairness, all baseline methods are evaluated using their official open-source implementations, default hyperparameter settings, and original configurations, with only the training datasets unified to our settings. Furthermore, all models are evaluated under completely identical hardware environments, data preprocessing pipelines, and standardized calculation scripts for PSNR and SSIM.

B. Quantitative Comparison Experiments

To evaluate the effectiveness of the proposed model, we select several representative state-of-the-art architectures for comparison. Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity (SSIM) are adopted as the quantitative metrics to evaluate restoration quality. Table I presents the objective metrics for different methods, where red, blue, and green indicate the best, second-best, and third-best performances, respectively.

The results demonstrate that, under identical hardware settings, our model delivers the most balanced and comprehensive performance. Although it does not achieve the absolute highest objective scores on individual dehazing and raindrop-removal tasks, it consistently attains highly competitive metrics across all tasks and ultimately achieves the highest average performance. In multi-task joint training, some All-in-One models tend to implicitly favor specific degradation distributions (e.g., haze) due to the inherent physical discrepancies and optimization conflicts among diverse weather effects, resulting in high peak scores on individual tasks but severe performance drops on others (such as snow removal). In contrast, our framework is designed to mitigate this architectural learning bias, prioritizing a robust feature representation that uniformly accommodates multiple degradations. Compared to models

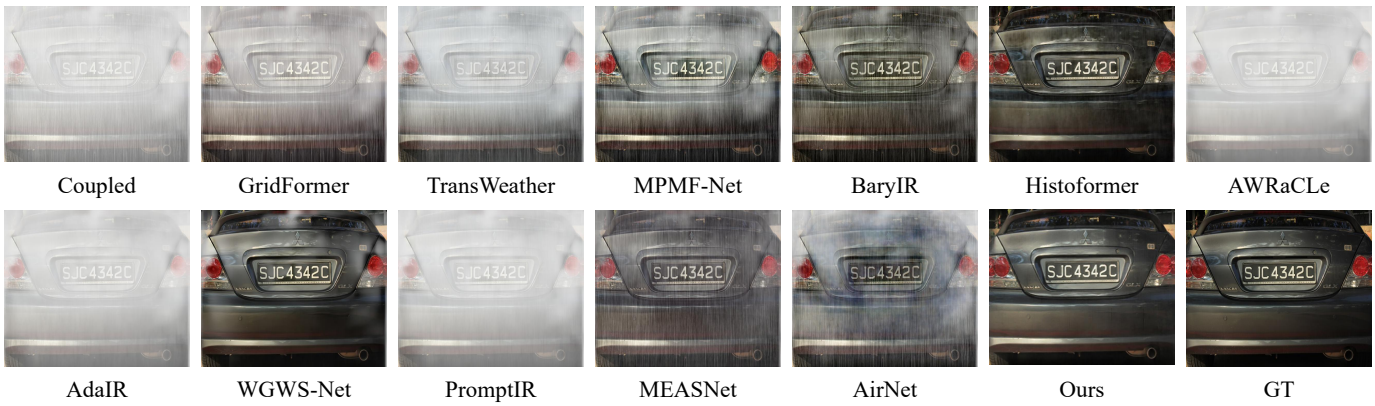


Fig. 9. Visual comparisons for generalization evaluation on the rain-haze coupled image dataset [61].

highly specialized for specific degradation types, our model exhibits superior versatility and multi-task generalization. Extensive experiments confirm that our model's scores are robust and balanced across tasks, indicating broad adaptability under diverse adverse weather conditions.

Furthermore, on the unseen rain-haze coupled dataset, while most comparison methods experience severe performance drops when confronted with intricately coupled degradations, our HPP-IR achieves the highest quantitative scores as shown in Table I. This remarkable zero-shot performance further validates that our two-level collaborative prompting paradigm effectively captures the intrinsic physical patterns of weather degradations rather than merely overfitting to the training distributions, thereby demonstrating exceptional robustness and generalizability in real-world coupled degradation scenarios.

C. Qualitative Comparison Experiments

In various degradation scenarios, our work demonstrates superior performance through its hierarchical progressive prompts and loss design, as shown in Fig. 5 to Fig. 8.

For dehazing, Histoformer and WGWS-Net suffer from under-restoration, while AdaIR, MEASNet, and AirNet tend to over-restore. By supervising the global and local intensity trends with the correlation-aware loss, our method avoids these severe intensity distortions, ensuring accurate and natural brightness recovery. Moreover, benefiting from the Degradation-Feature Prompt Module, our method accurately disentangles the haze layer and avoids visual artifacts, such as those observed in MEASNet and AirNet, by effectively mitigating the feature confusion and dilution commonly encountered in deep encoder architectures. In deraining tasks, the two-level prompting architecture thoroughly removes rain streaks while preserving critical structural details (e.g., locomotive components and stone arch bridge textures). In contrast, residual rain streaks and structural blurring remain visually obvious in the outputs of TransWeather, MPMF-Net, AWRaCLe, and AirNet. Additionally, our method maintains accurate color fidelity and preserves text details, which BaryIR lacks. In terms of raindrop removal, our approach effectively mitigates the conspicuous raindrop residues that commonly afflict TransWeather, MPMF-Net, AWRaCLe, and AirNet, thereby

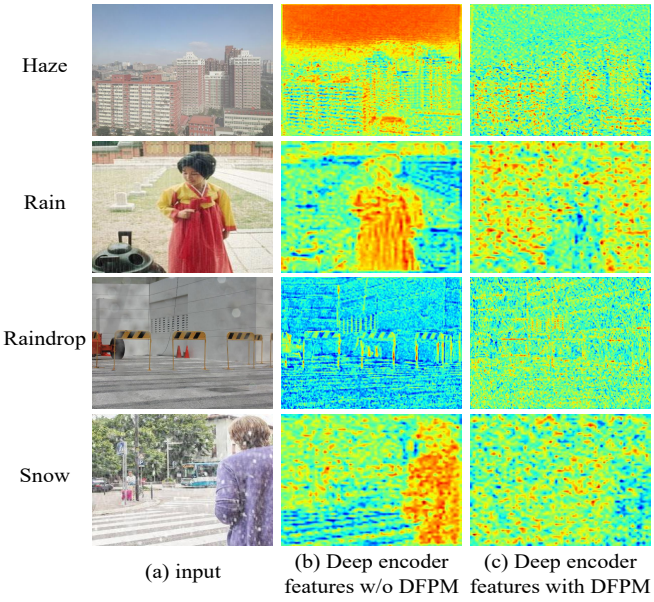


Fig. 10. Visual comparison of deep feature maps extracted from the deepest encoder layer.

yielding cleaner results with well-preserved text readability. For snow removal, our method achieves thorough restoration. Furthermore, benefiting from the DFPM module's ability to prevent degradation dilution and better separate degradation from the background, our model effectively overcomes severe feature confusion. While MPMF-Net, BaryIR, Histoformer, PromptIR, and MEASNet erroneously retain snow particles resembling the cloud background, our approach successfully removes them and faithfully preserves the intrinsic cloud textures of the wall paintings. To further demonstrate the model's generalization capability, we evaluate it on an unseen rain-haze coupled dataset. In this challenging scenario, most methods struggle to disentangle intricately coupled unseen degradations. WGWS-Net leaves partial residual degradation, indicating incomplete background-degradation separation—an issue effectively mitigated by our DFPM. Furthermore, benefiting from the Pearson loss function, our method maintains accurate and natural color fidelity, which Histoformer lacks.

Beyond external restoration performance, we visualize the intermediate feature representations to intuitively demonstrate the effectiveness of our DFPM in addressing feature dilu-

TABLE II
COMPARISON OF THE PARAMETERS, FLOPS, AND INFERENCE SPEED OF RESTORATION METHODS

Methods	Param(M)	FLOPs(GMac)	Inference Time(s)
MPRNet [53]	3.64	548.65	0.04
AirNet [18]	5.77	301.27	0.15
TransWeather [20]	38	6.13	0.02
WGWS-Net [54]	5.97	135.95	0.03
PromptIR [41]	32.97	158.14	0.10
GridFormer [55]	30.12	251.35	0.13
Histoformer [56]	16.62	91.59	0.08
MEASNet [50]	31.58	168.43	0.10
AdaIR [44]	28.73	147.18	0.09
AWRaCLe [57]	94.18	165.42	0.10
MPMF-Net [62]	3.30	30.86	0.03
BaryIR [59]	53.77	197.07	0.13
HPP-IR(Ours)	14.82	78.35	0.04

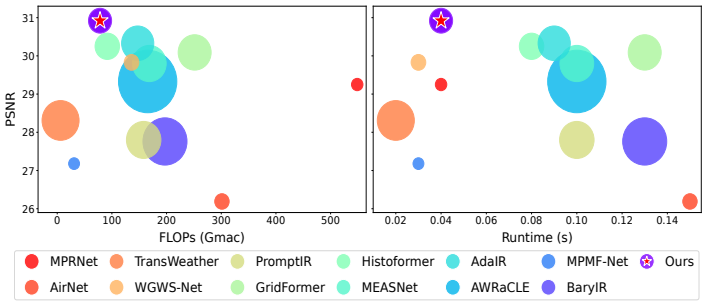


Fig. 11. Comparison of parameter count, computational cost, and inference speed for different image restoration methods. Dot size is proportional to model parameter magnitude.

tion and confusion. As visually demonstrated in Fig. 10, without explicit degradation guidance, the baseline model prioritizes intrinsic content information across all weather scenarios. It erroneously prioritizes macroscopic background structures—such as the depth-dependent outlines of skies and buildings (Haze), the glaring contours of foreground human figures (Rain and Snow), and rigid geometric edges like the high-contrast horizontal stripes of roadblocks (Raindrop). This disproportionate focus on abstract structural semantics overwhelms fine-grained weather cues, leading to severe feature dilution. In contrast, our proposed architecture, equipped with the DFPM, suppresses these semantic biases by steering deep feature activations to align with the physical distribution of degradations, ensuring robust preservation of degradation priors.

These results collectively confirm that the integrated design enables superior detail retention, thorough degradation removal, and robust cross-degradation generalization without task-specific adaptation.

D. Model Efficiency Comparison

The experimental results in Table II demonstrate that our proposed model maintains a relatively low parameter count and computational complexity compared to various state-of-the-art methods while still achieving excellent performance. This indicates that our method achieves a favorable balance between efficiency and effectiveness. As clearly illustrated in Fig. 11, compared to other methods, our model is highly suitable for deployment in practical image degradation processing systems with real-time requirements. Specifically, the reported inference time of 0.04s is evaluated on 256×256 input images. In addition, we conducted supplementary inference tests at a higher 512×512 resolution, which is more aligned with real-world scenarios, yielding a result of 0.15s.

Furthermore, to provide a transparent analysis of the module overheads, we quantify the total computational cost introduced by our complete prompting components. Our entire HPP-IR model contains 14.80 M parameters and requires 78.31 GMacs. The progressive deployment of DFPMs (from the second stage onward) introduces a cumulative overhead of merely 0.1107 M parameters (∼0.74% of the total network) and 0.2636 GMacs (∼0.34%), with an imperceptible latency increment of 0.406 ms. Similarly, all CLPC modules combined require only 0.1701 M parameters (∼1.15%) and 1.65 GMacs (∼2.11%), adding just 1.473 ms to the total inference time. This extremely marginal cumulative overhead brings a substantial 1.19 dB average PSNR improvement, indicating that our method achieves a highly favorable balance between efficiency and effectiveness.

TABLE III
EFFECT OF THE PROPOSED PLUG-IN MODULES ON IMAGE RESTORATION PERFORMANCE

Designs	M1	M2	M3	M4	M5 (Ours)
DFPM	✓	✗	✓	✓	✓
CLPC	✓	✓	✗	✓	✓
add $\mathcal{L}_{cor}$	✗	✓	✓	✓	✓
Prompt Collab.	✓	–	–	✗	✓
SOTS	27.95/0.9163	28.65/0.9235	27.94/0.9314	26.40/0.9318	**29.32/0.9321**
Rain1400	32.04/0.9156	32.48/0.9224	32.31/0.9228	**33.44**/0.9388	33.19/**0.9510**
Raindrop	29.71/0.9221	30.57/0.9433	30.88/0.9326	**31.47**/0.9316	31.01/**0.9465**
Snow100K	29.20/0.9034	29.63/0.9251	29.81/0.9134	29.27/0.9083	**30.15/0.9355**
Average	29.73/0.9144	30.33/0.9286	30.24/0.9251	30.15/0.9276	**30.92/0.9413**

TABLE IV
QUANTITATIVE ASSESSMENT OF SPATIAL CORRELATION BETWEEN DEEP FEATURES AND DEGRADATION RESIDUALS

	Haze	Rain	Raindrop	Snow	Average
Baseline (M2)	-0.1003	-0.0596	0.0925	-0.0116	-0.0197
Ours (DFPM)	0.4624	0.0527	0.1491	0.1066	0.1927
Improvement	**+0.5627**	**+0.1123**	**+0.0566**	**+0.1183**	**+0.2125**

E. Ablation Study

To verify the proposed components, ablation studies are presented in Table III. Removing any single module (M1-M3) degrades performance, confirming the individual necessity of DFPM, CLPC, and correlation-aware loss. Crucially, we introduce a separated variant (M4) where DFPM and CLPC operate independently. While M4 achieves high peak scores on specific tasks, it suffers a severe performance collapse in dehazing, illustrating the inherent optimization conflicts

TABLE V
ABLATION STUDY ON THE INTERNAL STRUCTURE OF DFPM

Designs	SOTS	Rain1400	Raindrop	Snow100K	Average
Baseline (M2)	28.65/0.9235	32.48/0.9224	30.57/0.9433	29.63/0.9251	30.33/0.9286
Only SE	26.36/0.9461	33.26/0.9364	31.57/0.9313	29.17/0.9068	30.09/0.9301
SE+1×1	27.59/**0.9529**	**33.43**/0.9385	**31.58**/0.9316	29.25/0.9079	30.47/0.9327
SE+3×3	**29.32**/0.9321	33.19/**0.9510**	31.01/**0.9465**	**30.15/0.9355**	**30.92/0.9413**
SE+5×5	27.94/0.9527	33.29/0.9368	31.42/0.9314	29.19/0.9070	30.46/0.9320

TABLE VI
PERFORMANCE AND COMPUTATIONAL OVERHEAD TRADE-OFFS FOR DIFFERENT DFPM INSERTION STAGES

Insertion Strategy	Δ Param (M)	Δ FLOPs (GMac)	Δ Time (ms)	Average PSNR / SSIM
Baseline (M2)	0	0	0	30.33 / 0.9286
Only First Stage	+0.0015	+0.0983	+0.127	30.42 / 0.9316
From 2nd Stage (Ours)	+0.1107	+0.2636	+0.406	**30.92 / 0.9413**
All Stages	+0.1122	+0.3619	+0.533	30.74 / 0.9321

in all-in-one restoration when modules work in isolation. In contrast, our full model (M5) integrates these modules via hierarchical collaborative prompting. This synergistic architecture effectively resolves feature conflicts between diverse weather effects, mitigating task-specific optimization bias and achieving the highest overall average performance.

Furthermore, to quantitatively validate that DFPM effectively addresses the feature dilution issue, we evaluate the Pearson Correlation Coefficient (PCC) between the deep latent features and the true physical degradation residuals in Table IV. Note that the baseline network without DFPM is denoted as Baseline (M2), corresponding exactly to the M2 configuration in Table III. This specific notation is consistently applied across all subsequent ablation tables. For a fair evaluation, both Baseline (M2) and our model are trained with the same joint loss ($\mathcal{L}_{rec} + \alpha\mathcal{L}_{cor}$), ensuring the PCC metric solely reflects the architectural feature extraction capabilities. A negative or near-zero PCC for Baseline (M2) indicates severe feature dilution and misalignment with true degradation, demonstrating that this configuration overfits to background semantics. In contrast, our DFPM-equipped model achieves positive PCC values across all four weather conditions, delivering a significant average improvement of +0.2125. These positive values demonstrate the strong spatial anchoring of degradation priors, mathematically verifying that DFPM successfully forces the network to focus on pure degradation statistics.

To investigate the internal structural design of DFPM, we evaluate variants with different receptive fields in Table V. The baseline model (w/o DFPM) yields an average PSNR of 30.33 dB. Replacing it with pure channel attention (Only SE) causes the PSNR to drop to 30.09 dB. This performance degradation indicates that the global pooling in standard SE struggles to preserve fine-grained, spatially dependent degradation features (e.g., rain and snow), thus disrupting the original feature flow. To mitigate this spatial blindness, we introduce convolutions with varying kernel sizes. As observed, the SE+1×1 variant still lacks sufficient spatial context, while expanding the receptive field to SE+5×5 introduces redundant background noise. Ultimately, our SE+3×3 design strikes the optimal balance, achieving a peak performance of 30.92 dB. This proves that a 3×3 convolution is strictly necessary to aggregate fine-grained spatial context, ensuring the extraction of complete degradation priors to synergize with the subsequent CLPC module.

To determine the optimal deployment strategy for DFPM, we evaluate different insertion stages in Table VI. Applying DFPM solely at the first encoder stage yields a marginal PSNR improvement of only 0.09 dB over the baseline, confirming that the initial layer already possesses intact degradation cues; additional prior injection here primarily introduces computational redundancy. Conversely, deploying DFPM across all stages causes a noticeable performance drop, as excessive degradation priors in shallow layers interfere with the normal extraction of background semantics. Striking the optimal balance, our default strategy applies DFPM from the second stage onward. This configuration achieves the peak restoration capability while strictly limiting the overall parameter increment to merely 0.1107 M and the latency to 0.406 ms. This progressive design effectively mitigates deep-layer feature dilution while preventing shallow-level interference, demonstrating high suitability for real-time deployment.

To empirically justify our design decisions regarding the loss function, Table VII provides a comprehensive evaluation comparing the introduced correlation-aware loss ($\mathcal{L}_{cor}$) against other commonly used structural constraints, namely the SSIM loss ($\mathcal{L}_{SSIM}$) and the Perceptual loss ($\mathcal{L}_{perc}$). As theoretically analyzed in Sec. III-C, $\mathcal{L}_{SSIM}$ is sub-optimal for handling severe, globally distributed intensity shifts, yielding lower average performance. Similarly, $\mathcal{L}_{perc}$ provides competitive pixel-wise recovery but yields an inferior structural similarity, alongside the aforementioned heavy training memory footprint. In stark contrast, our PCC-based $\mathcal{L}_{cor}$ is a completely parameter-free statistical metric. By explicitly enforcing intrinsic linear intensity correlations, it elegantly achieves the highest comprehensive visual quality (30.92 dB in PSNR and 0.9413 in SSIM) across multi-weather scenarios without introducing any training or inference burden.

Additionally, we systematically explore the impact of the correlation loss weight α by varying it from 0.1 to 10. As shown in Table VII, the performance exhibits a clear inverted U-shaped trajectory, demonstrating that the optimal structural-pixel balance is achieved at $\alpha = 1$. Lower weights (e.g., $\alpha \leq 0.5$) provide insufficient structural constraint, limiting the restoration of local details. Conversely, excessively high weights overly restrict the pixel distribution optimization space, hindering pixel-wise accuracy and gradually degrading the overall performance. Consequently, we set $\alpha = 1$ as the robust default configuration to ensure consistent performance improvement without requiring task-specific parameter tuning.

V. CONCLUSION

In this study, we propose a progressive two-level prompting architecture aimed at addressing key challenges in all-in-one

TABLE VII
ABLATION STUDY ON DIFFERENT LOSS FUNCTIONS

Designs	α	SOTS		Rain1400		Raindrop		Snow100K		Average	
		PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
w/o $\mathcal{L}_{\text{cor}}$	0	27.95	0.9163	32.04	0.9156	29.71	0.9221	29.20	0.9034	29.73	0.9144
with $\mathcal{L}_{\text{cor}}$	0.1	27.67	0.9266	32.07	0.9163	29.40	0.9164	29.17	0.9076	29.58	0.9167
	0.5	27.98	0.9534	32.89	0.9381	31.18	0.9293	28.56	0.8931	30.15	0.9285
	1	**29.32**	0.9321	**33.19**	**0.9510**	31.01	**0.9465**	30.15	**0.9355**	**30.92**	**0.9413**
	5	28.08	0.9301	32.62	0.9256	31.26	0.9395	**30.16**	0.9346	30.53	0.9325
	10	27.86	0.9267	32.58	0.9246	**31.31**	0.9217	29.96	0.9141	30.43	0.9218
with $\mathcal{L}_{\text{SSIM}}$	0.1	28.42	**0.9546**	32.74	0.9361	30.76	0.9255	28.42	0.8911	30.09	0.9268
with $\mathcal{L}_{\text{perc}}$	0.1	28.79	0.9543	33.15	0.9347	30.78	0.9251	28.78	0.9030	30.38	0.9293

image restoration under diverse weather degradation factors. Specifically, our Degradation-Feature Prompt Module explicitly captures and preserves original degradation priors via parallel prompting branches at the encoder stage, effectively preventing feature dilution and facilitating accurate disentanglement. Subsequently, the Cross-Layer Prompt Component receives these first-level degradation prompts and injects context-aware restorative guidance at the skip connections between the encoder and decoder, significantly enhancing the specificity and adaptability of feature transfer. Furthermore, by incorporating correlation-aware pixel-level constraints, our method explicitly enforces intrinsic structural consistency between the restored and ground truth images, leading to substantial improvements in overall visual quality. Extensive experimental results demonstrate that the proposed HPP-IR framework exhibits outstanding performance on various weather datasets. It validates exceptional effectiveness and robustness in complex degradation scenarios, enabling high-quality image data for downstream tasks while maintaining highly efficient computational performance.

However, the performance of the proposed model under extreme degradation conditions still has the potential for further improvement, which can be achieved by enhancing the modeling of physical scattering effects and local distortions, as well as by optimizing the inherent learning bias in multi-task training to enable the model to focus on a broader range of degradation patterns. Furthermore, as a preliminary exploration of the correlation-aware pixel-level constraint in all-in-one restoration, we also plan to deeply investigate the adaptive improvements of the PCC loss (e.g., incorporating local windows or multi-scale computations) and validate its generalizability across other mainstream architectures. These directions will be the focus of our future research.

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